

# Theoretical Neuroscience : Chapter 2

## Neural Encoding II: Reverse Correlation and Visual Receptive Fields

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- **Neurons** propagate signals rapidly over large distance;
  - they do this generating characteristic electrical pulse called **spikes**;
  - they transmit informations by **firing sequences of spikes**;
- **Neural coding** deals with measuring and characterizing how stimulus attributes are represented by action potentials (spikes);
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*predict responses from stimuli*
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- Neurons typically respond by producing complex spike sequences;
  - complexity + trial-to-trial variability  $\simeq$  non-deterministic model;
  - **probabilistic approach**: *probabilities that different spike sequences are evoked by a specific stimulus*;
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  - [rtc] **neural response tuning curve**<sup>2</sup>, average firing rate written as function of a stimulus,  $\langle r \rangle = f(s)$ ;
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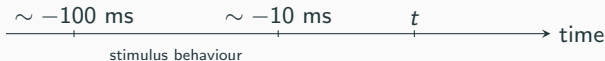
**Reverse correlation methods** can be used to construct a more accurate model that includes the effect of the stimulus over an extended period time.

The simplest way to construct an estimate of the firing rate  $r(t)$  evoked by a stimulus  $s(t)$  is to assume that *the firing rate at any given time  $t$  can be expressed as a weighted sum of the values taken by the stimulus at earlier times:*

$$r_{est}(t) = r_0 + \int_0^{\infty} D(\tau)s(t - \tau)d\tau$$

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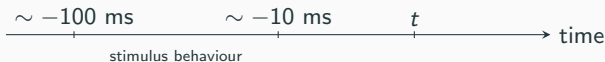
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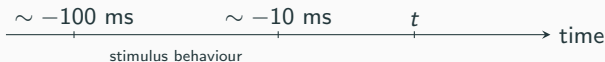
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## Heuristic justification: Volterra series (non-linearity + memory effect)

- The **Volterra series**

$$u(t) = k_0 + \sum_{n=1}^{\infty} \frac{1}{n!} \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} k_n(\tau_1, \dots, \tau_n) \prod_{j=1}^n v(t - \tau_j) d\tau_j,$$

$k_n$  is the **Volterra kernel** of order  $n$ , is a model for nonlinear behaviour similar to the **Taylor series**

$$y(t) = f(a) + \sum_{n=1}^{\infty} \frac{f^{(n)}(a)}{n!} (t - a)^n.$$

- It differs from the Taylor series in its ability to capture both non-linear and *memory* effects;
  - It can be considered a nonlinear generalization of the convolution operator;
  - **Taylor** approximate the response of a nonlinear system to a given input if the output depends strictly on the input at that *particular time*;
  - in **Volterra** the output of the nonlinear system depends on the input to the system at *all other times*.

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To *construct an estimate of the firing rate* we choose the kernel  $D$  to *minimize* the squared difference between the estimated response to a stimulus and the actual measured average over the duration of the trial,  $T$ :

$$E = \frac{1}{T} \int_0^T (r_{\text{est}}(t) - r(t))^2 dt.$$

whose minimum is

$$\int_0^\infty D(\tau') \left( \frac{1}{T} \int_0^T s(t - \tau') s(t - \tau) dt \right) d\tau' = \frac{1}{T} \int_0^T (r(t) - r_0) s(t - \tau) dt.$$

The result is that  $D$  satisfies an equation involving two quantities introduced in Chapter 1: the **firing rate-stimulus correlation function**  $Q_{rs}(-\tau)$  and the **stimulus autocorrelation function**  $Q_{ss}(\tau - \tau')$ .

The method we are describing is called **reverse correlation** because the firing rate-stimulus correlation function is evaluated at  $-\tau$ .

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## Prediction of the firing rate

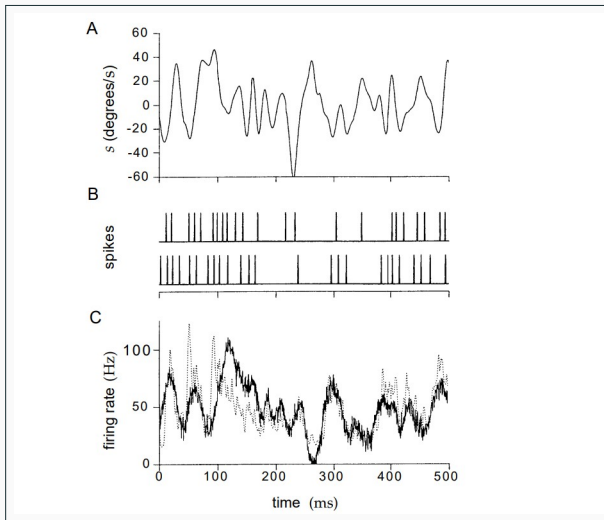


Figure 2.1, page 48.



## Static non-linearities

- The optimal kernel produces an estimate of the firing rate that is a **linear** function of the stimulus. Problems:
  - nothing prevent the predicted  $\hat{r}$  from becoming negative;
  - the predicted rate increases without a bound as the magnitude of the stimulus increases.
- However, neurons and nervous system are **nonlinear**, so a linear estimate is an *approximation*.

One way to overcome the deficiencies of a linear prediction is to write the firing rate as a background rate pulse as a nonlinear function of the linearly filtered stimulus. Let

$$L(t) := \int_0^\infty D(\tau)s(t-\tau)d\tau.$$

Then, the modification is to replace the linear prediction  $r_{\text{est}}(t) = r_0 + L(t)$  with the generalization

$$r_{\text{est}}(t) = r_0 + F(L(t)),$$

with  $F$  arbitrary function called **static non-linearity** (can be extracted from data). If  $F$  properly bounded the estimated  $\hat{r}$  will never be negative or unrealistically large.

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- Static non-linearities are restrictive:
  - no dependence on weighted autocorrelations;
  - no dependence on higher-order terms in the Volterra series:
- No optimality conditions on the kernel;
  - a Theorem by Bussang, suggest that  $r_{est}(t) = r_0 + F(L(t))$  will provide a reasonable kernel even in presence of a static non-linearity if the white noise stimulus used is Gaussian.

In some cases, the linear term of the Volterra series fails to predict the response even when static non-linearity are included;

- The accuracy can sometimes be improved by replacing the parameter  $s$  with an appropriate  $f(s)$ :

$$L(t) = \int_0^{\infty} D(\tau) f(s(t - \tau)) d\tau,$$

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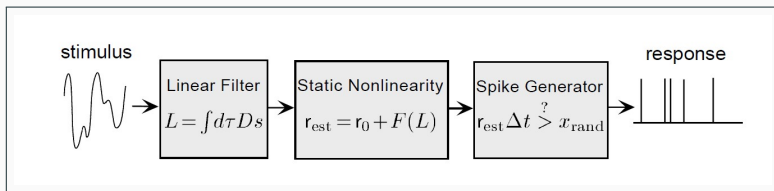


Figure 2.3, page 51.

- integral of the stimulus times the optimal kernel is computed;
- the estimated firing rate is background + nonlinear function of the linear filter;
- the estimated firing rate is used to drive a Poisson process that generates spikes.
  - the expected number of spikes  $\lambda$  in a time step  $\Delta t$  is related to the rate  $r_{\text{est}}$  as  $\lambda = r_{\text{est}} \Delta t$ ;
- a simpler way of modelling a Poisson process is to choose whether or not there is an event for each time step; this can be done by drawing a random number  $x_{\text{rand}} \in [0, 1]$ ; if  $r_{\text{est}} \Delta t > x_{\text{rand}}$  the spike is fired, otherwise it is not.

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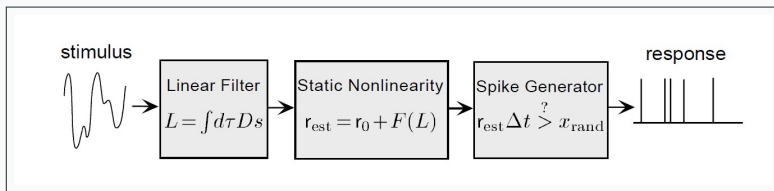


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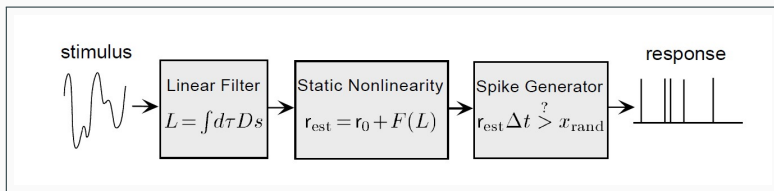


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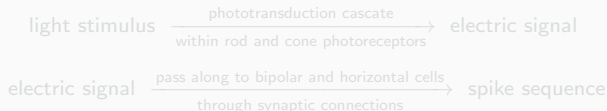
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## Early stages of the visual system: basic anatomy and physiology

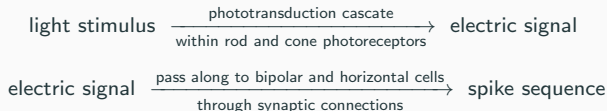
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- some basics of the **early stages of the visual system**;
  - First stop: **retina**;



- Second stop: **thalamus**;  
the output neurons of the retina are the **retinal ganglion cells**, whose axons form the **optic nerve** which conducts the output spike trains of the retinal ganglion to the **lateral geniculate nucleus (LGN)** of the thalamus.
- Last stop: **primary visual cortex** (passing through **optic chiasm**).

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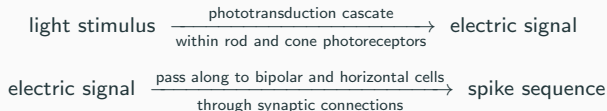
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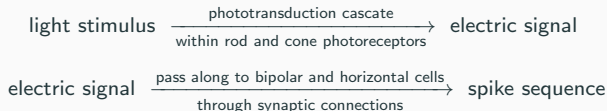
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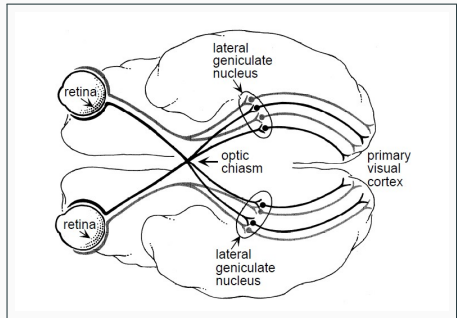
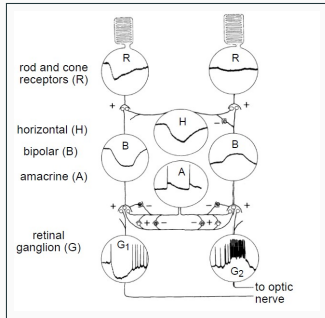
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# Early stages of the visual system: basic anatomy and physiology



Figures 2.4/2.5, page 52/53.

- The above system respond to light stimuli in restricted regions of the visual field called their **receptive fields**;
  - Within the receptive fields, there are regions where illumination higher than the background light intensity enhances firing, and other regions where lower illumination enhances firing;
  - The spatial arrangement of these regions determines the selectivity of the neuron to different inputs.
- Visually responsive neurons in the retina, LGN, and primary visual cortex are divided into **two** classes, depending on whether or not the contributions from different locations within the visual field sum linearly:
  - **Simple cells**;
  - **Complex cells** (non-linearities must be included in descriptions of their responses).
- A first step in studying the selectivity of any neuron is to identify the types of stimuli that evoke strong responses.

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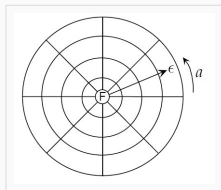
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## The retinotopic map

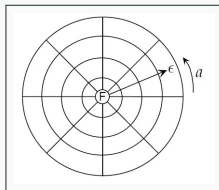
- the visual world is mapped onto the cortical surface in a topographic manner;
- the **retinotopic map** refers to the transformation from the coordinates of the visual world to the corresponding locations on the cortical surface;
- objects located a fixed distance from one eye lie on a sphere;
  - in this system of coordinates, the latitude coordinate is called the eccentricity,  $\epsilon$ , and the longitude coordinate, measured from the horizontal meridian, is called the azimuth,  $a$ .



Figures 2.6, page 55.

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Figures 2.6, page 55.

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$$s = s(t)$$

to characterize a time-dependent stimulus.

- But, the description of visual stimuli is actually more complex.
- We describe the stimulus by a function

$$s = s(x, y, t)$$

that is proportional to the difference between the luminescence at a point  $(x, y)$  at time  $t$  and the average or background level of illumination.

- During recordings, visual neurons are usually stimulated by images that vary over both space and time:

$$s(x, y, t) = A \cos(Kx \cos \Theta + Ky \sin \Theta - \Phi) \cos(\omega t),$$

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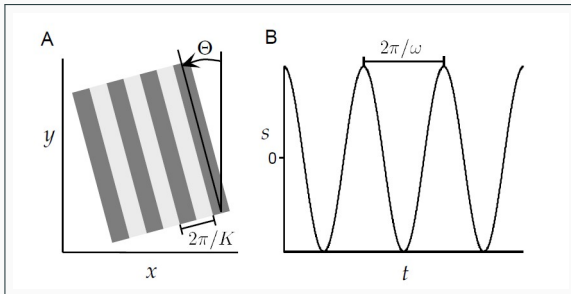
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**Reverse-Correlation Methods: Simple Cells**

Static Non-linearities: Complex Cells

[sta] **spike-triggered average**, answer to the question: "What, on average, did the stimulus do before an action potential was fired?",  $\langle s \rangle$ .

- [sta] will be functions of three variables

$$C(x, y, t) = \frac{1}{\langle n \rangle} \left\langle \sum_{i=1}^n s(x, y, t_i - \tau) \right\rangle$$

and represents the average value<sup>4</sup> of the visual stimulus at  $(x, y)$  a time  $\tau$  before the spike was fired. We can define the correlation function between firing rate, a time  $t$  and stimulus at time  $t + \tau$

$$Q_{rs}(x, y, \tau) = \frac{1}{T} \int_0^T dt r(t) s(x, y, t + \tau).$$

Relation between spike-triggered average and reverse-correlation function<sup>5</sup>

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- To estimate the firing rate of a neuron in response to a particular image, we add a function of the output of a linear filter of the stimulus to the background firing rate

$$r_{est}(t) = r_0 + F(L(t)),$$

$L(t)$  is obtained by integrating over the past history of the stimulus with a kernel acting as a weight function (see slide 8).

- Visual stimuli depend on the spatial location, so we must decide how contributions from different image locations are to be combined to determine  $L(s)$ .
- The **simplest assumption** is that the contribution from different spatial points sum linearly

$$L(t) = \int_0^\infty d\tau \int dx dy D(x, y, \tau) s(x, y, t - \tau).$$

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# Spatial receptive fields

- Spatial structure of spike-triggered average stimuli for two simple cells in the primary visual cortex of a cat.
  - where  $D_s$  is positive, we call it **ON region**;
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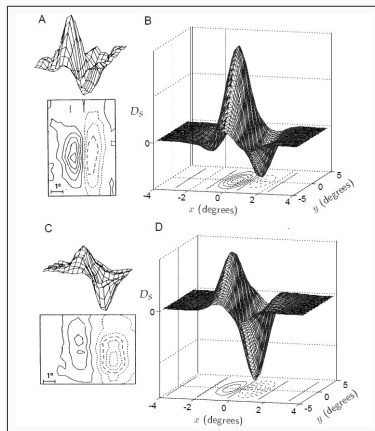


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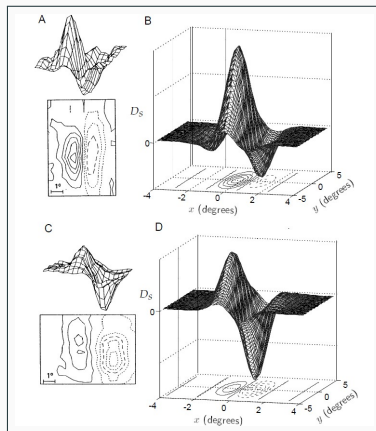


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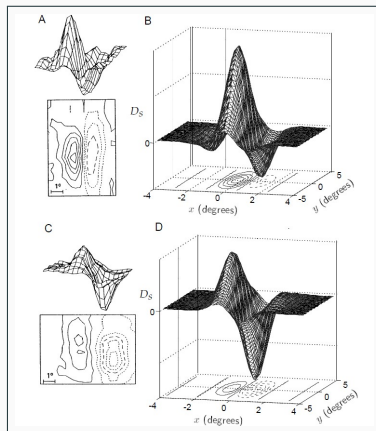


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- A mathematical approximation of the spatial receptive field of a simple cell is provided by a **Gabor function**:

$$D_s(x, y) = \frac{1}{2\pi\sigma_x\sigma_y} \exp\left(-\frac{x^2}{2\sigma_x^2} - \frac{y^2}{2\sigma_y^2}\right) \cos(kx - \phi).$$

- These are the only functions used to fit spatial receptive fields of simple cells.
- The parameter of this function determine the properties of the spatial receptive field:
  - $\sigma_x$  and  $\sigma_y$  extent in  $x$  and  $y$  direction,
  - $k$  preferred spatial frequency<sup>6</sup>
  - $\phi$  preferred spatial phase<sup>7</sup>

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- There are experimental evidences that neurons has, to a good approximation, a separable space-time receptive field.
- When a space-time receptive field is separable the reversal can be described by a function (Aldeson and Berger, 1985)

$$D_t(\tau) = \alpha \exp(\alpha\tau) \left( \frac{(\alpha\tau)^5}{5!} - \frac{(\alpha\tau)^7}{7!} \right), \quad \tau \geq 0,$$

$D_t(\tau) = 0$  for  $\tau < 0$ ,  $\alpha$  is a constant that sets the scale for the temporal development.

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Now the **question** is: *"What is the response of a simple cell to a counterphase grating stimulus?"*.

- This can be estimated by computing the function  $L(t)$ .
- In case of separability the linear estimate of the response can be written as the product of two terms:

$$L(t) = L_s L_t(t),$$

$$L_s(t) = \int dx dy D_s(x, y) A \cos(Kx \cos \Theta + Ky \sin \Theta - \Phi),$$

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- Many neurons in primary visual cortex are selective for the direction of motion of an image.
- Accounting for direction selectivity requires **non-separable** space-time receptive fields.
- How to describe a non-separable receptive field structure?
  - Use a separable function constructed from a product of a Gabor function for  $D_s$  and  $D_t$  as we defined before.
  - Then, write these functions as mixture of rotation of the  $x$  and  $\tau$  variable:

$$D(x, y, \tau) = D_s(x', y)D_t(\tau'),$$

with

$$x' = x \cos(\psi) - c\tau \sin(\psi),$$

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$$r_{est}(t) = r_0 + F(L(t)),$$

where  $F$  is an appropriate chosen static non-linearity.

- Simplest choice: rectification of the fire rate

$$F(L) = G[L]_+$$

where  $G$  set to fit the magnitude of the measured firing rates.

- This choice **does not match with the data!**
- This led Haager (1992) to propose that an appropriate static non linearity to use is

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**Static Non-linearities: Complex Cells**

- **Linear methods** (i.e. spike-triggered average), are useful for revealing the properties of simple cell;
- Complex cells display features that are fundamentally incompatible with a linear description;
  - the spatial receptive fields of complex cells cannot be divided into separate **ON** and **OFF** regions that sum linearly to generate the response.
- Comparison
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## Receptive fields in the retina and LGN

- In conclusion we return to the initial stages of the visual pathway and briefly describe the receptive field properties of neurons in the retina and LGN.
- Retinal ganglion cells display a wide variety of response characteristics, including nonlinear and direction-selective responses.
- The spatial structure of the receptive fields of these neurons has a center-surround structure consisting either of a circular central ON region surrounded by an annular OFF region, or the opposite arrangement of a central OFF region surrounded by an ON region.

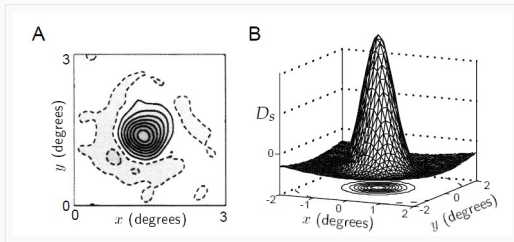


Figure 2.25, page 78.

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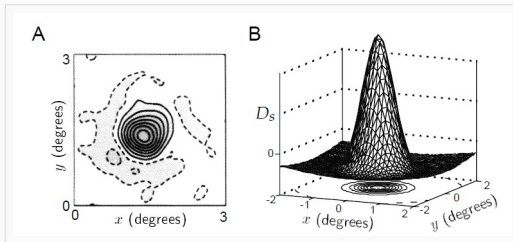


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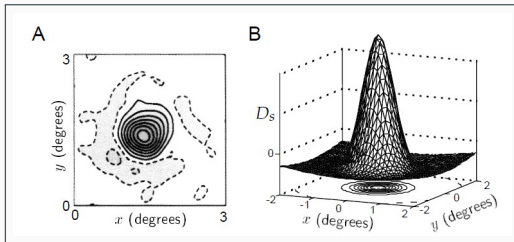


Figure 2.25, page 78.

- The spatial structure of retinal ganglion and LGN receptive fields is well captured by a difference-of-Gaussians model in which the spatial receptive field is expressed as

$$D_s(x, y) = \pm \left( \frac{1}{2\pi\sigma_{cen}^2} \exp\left(-\frac{x^2 + y^2}{2\sigma_{cen}^2}\right) \right) - \frac{B}{2\pi\sigma_{sur}^2} \exp\left(-\frac{x^2 + y^2}{2\sigma_{sur}^2}\right)$$

Here the center of the receptive field has been placed at  $x = y = 0$ .

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# Comparison of predicted and measured firing rates

Cat LGN neuron responding to a videomovie:

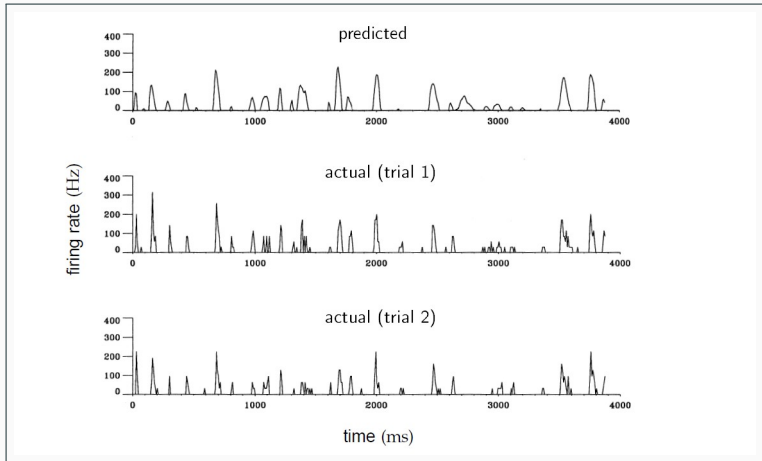


Figure 2.26, page 79.

- The models of visual receptive fields we have been discussing are purely **descriptive**, but they provide an important framework for studying how the circuits of the retina, LGN, and primary visual cortex generate neural responses.
- There are examples of a more **mechanistic** model, e.g. Hubel and Wiesel in 1962 proposed that the oriented receptive fields of cortical neurons could be generated by summing the input from appropriately selected LGN neurons.

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Thanks for your attention!